

PLANNING IN MATHEMATICS

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6.1 INTRODUCTION

Teaching is a process that involves four steps, namely Planning, Delivering the content matter or Transaction, Evaluation and Remediation. Teaching in a class where learners of different capabilities, interests, caste, creed, language, culture, etc., sit together makes it a very challenging task for the teacher to make the process of teaching effective and successful for each of the learners. The same is possible only through effective planning. The effective planning of teaching includes: What to teach? When to teach? How to teach? The answers to these questions require annual, monthly and daily planning. For such planning, you should know the curricular goals of mathematics for a particular class, learning outcomes to be achieved for each unit, activities to be performed for concept development, learning resources to be used, strategies to be adopted for



sustaining learners' interest, and evaluation techniques to be used. It means that planning has to be done at each and every step, which ranges from one lesson to the whole curriculum of mathematics for each class. In this chapter, we will discuss the importance of planning, curricular goals, learning outcomes, unit planning and lesson planning in mathematics at the foundational and preparatory stages of school education.



6.2 LEARNING OUTCOMES

After completion of the chapter, the learner:

- designs curricular goals and learning outcomes;
- explains the importance of planning; and
- develops a unit plan and daily plan for mathematics learning.

6.3 CURRICULAR GOALS AND LEARNING OUTCOMES

6.3.1 Curricular Goals

Curricular goals are the statements about a course or a subject that tell us broadly about the goals and implications of that course or subject. Curricular goals are derived from the philosophy, approaches emphasized in the National Curriculum Framework (NCF) developed from time to time. From the approaches or goals suggested in NCFs, the curricular goals of the specific course or subject are derived. In general terms we can say that curricular goals help educationists, teachers and learners in understanding the 'what' and 'why' of a subject. Directions are given to curriculum developers, enabling them to align educational aims by providing a framework for teaching-learning strategies and facilitating assessment.

6.2.2 Importance Of Curricular Goals

The curricular goals of a course or a subject are very important for knowing the 'what' and 'why' of the course. These goals guide the purpose of any course for teachers as well as learners. The teachers come to know about learning outcomes, instructional strategies, and assessment techniques to be adopted and learners come to know about expectations from them after completing the course. In a way, curricular goals give direction to all stakeholders of that course. We can say that curricular goals are important, as they:

- have a definite purpose.
- tell us about the intended aims of a course or a subject.
- guide curriculum developers in developing instructions and assessment techniques.
- are framed by aligning broader aims of education with the nature of that subject or course.
- form the basis for designing any curriculum.
- help in assessing the overall effectiveness of a course or a subject on its completion for its further improvement.
- provide a road map for teachers and learners about the learning outcomes of a course or a subject.



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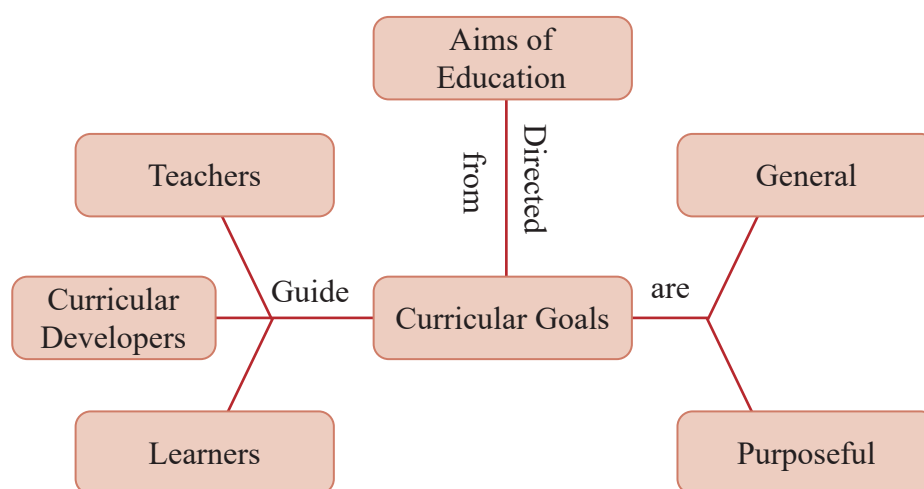


Fig 6.1: Concept Map of Curricular Goals

6.3.3 Writing of Curricular Goals

The curricular goals should be observable and should specify what learners would be able to perform or apply after completing the curriculum. Active verbs are used to describe the action we want learners to exhibit. Goals are aligned with the nature of the subject. It is ensured that goals are feasible and achievable for the majority of learners. Let us discuss the writing of curricular goals through five steps:

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Table 6.1

Step 1.	Purpose	While writing the purpose of curricular goals, three questions are answered: • Whether the curricular goals are of a course or a subject or a unit or a lesson? • What is the time duration of the course or subject or a unit or a lesson? • What are the learning outcomes of that course or subject or a unit or a lesson?
Step 2.	Expectations from learners	While writing the expectations from learners, three questions are answered: • What will learners be able to do at the end of the course or subject or a unit or a lesson? • What learning outcomes will learners achieve at the end in terms of levels of Bloom's Taxonomy? • What are the important competencies that learners need to acquire at the end of the course or subject or a unit or a lesson?
Step 3.	Specific Objectives	While writing specific objectives, again two questions are answered: • Whether verbs used describe the measurable behavior of the learner? • Whether the language used to write objectives is clear and free from any ambiguity?
Step 4.	Achievability of Objectives	While writing about the achievability of objectives, again three questions are answered: • Do the objectives align with the standard of the curriculum? • Are the objectives designed to be feasible for a majority of learners? • Do the learners have resources for the achievability of the objectives framed?
Step 5.	Reflection of curricular goals	Learners are assessed through formative and flexible techniques to check their progress at regular intervals. Curricular goals are reflected upon to check their relevance to the needs of the learner.

Writing of Specific Objectives:

‘Learners understand a triangle’ is a vague objective, as it is not specific. How is the understanding of a learner assessed?

Whereas the clear objective may be any of the following:

- The learner is able to identify a triangle.
- The learner is able to draw a triangle.
- The learner is able to define a triangle.
- The learner is able to classify a triangle.
- The learner is able to name a triangle.

Bloom’s Taxonomy is a hierarchical framework used to frame learning objectives in measurable terms. It was developed by educational psychologist Benjamin Bloom and his committee in the 1950s and later revised. The taxonomy is divided into three domains: cognitive (thinking), affective (feeling), and psychomotor (physical/manual). The cognitive domain, which is the most widely used, is further divided into six levels: Remember, Understand, Apply, Analyse, Evaluate, and Create. Action verbs classified under each domain are as follows:

A. Cognitive Domain (Thinking)**1. Remember:**

Recalling facts, basic concepts, and information.

- Examples: Reciting a table, recalling the formula for the area of a rectangle, defining a triangle, etc.
- Key verbs: Define, describe, identify, list, recall, and recognize.

2. Understand:

Explaining ideas or concepts.

- Examples: Explaining the difference between a square and a rectangle, etc.
- Key verbs: Explain, interpret, summarise, translate, simplify.

3. Apply:

Using learned information in a new situation.

- Examples: Using a formula to solve word problems.
- Key verbs: apply, calculate, demonstrate, solve, and use.



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**4. Analyse:**

Breaking down information into parts and understanding the relationships between them.

- Examples: Identifying what is given and what is to be found in the given problem.
- Key verbs: Analyse, compare, differentiate, distinguish, and examine.

5. Evaluate:

Making judgments based on criteria and standards.

- Examples: Evaluating the solution of a problem.
- Key verbs: assess, critique, evaluate, justify, support, and recommend.

6. Create:

Producing something new by combining elements in a novel way.

- Examples: Designing a new product, writing a new story, and formulating a new hypothesis.
- Key verbs: build, create, design and develop.

B. Affective Domain (Feeling)

This domain focuses on attitudes, values, interests, and emotions. It includes receiving information, responding to it, valuing it, organising it, and internalising it.

C. Psychomotor Domain (Physical/Manual)

This domain deals with physical skills and abilities. It involves imitation, manipulation, precision, articulation, and naturalisation.

Purpose and Application:

Bloom's Taxonomy is used to develop learning objectives, design instructional material, assess learning and promote higher-order thinking skills in learners.

Note:

- Learning activities and curricular goals are not the same, as learning activities are a way to achieve learning goals, while with curricular goals, the learners' previous knowledge and their learning environment should be given due weightage.

**CHECK YOUR PROGRESS 6.1**

A teacher of class 4th designed an objective: “Learners will understand the concept of triangle.” What suggestions would you give to her to improve upon her learning objectives? Also write any two learning objectives for the concept of triangle.

6.3.4 Examples of Curricular Goals in Mathematics

We have a few curricular goals in mathematics at the preparatory stage of school education as follows:

- The learners are able to conduct four fundamental operations of addition, subtraction, multiplication and division fluently.
- The learners are able to solve problems based on four fundamental operations, including reasoning and analysing the answer accordingly.
- The learners are able to explain numbers, measurement, data handling, patterns and shapes in their simple language.
- The learners are able to gain a positive attitude towards mathematics through joyful, engaging and hands-on activities.
- The learners are able to apply mathematical knowledge in real-life situations whenever needed.
- The learners are able to develop critical thinking skills through mathematical concepts.

These curricular goals let learners acquire fundamental skills for learning mathematics at higher stages by laying strong foundations of the subject.

**ACTIVITY 6.1**

1. Frame three curricular goals in mathematics at the foundational stage.
2. Compare the curricular goals of the foundational stage and the preparatory stage. What do you observe about the two?

6.3.5 Learning Outcomes

Learning outcomes are precise, focused, and measurable statements that define what learners are expected to achieve or demonstrate by the end of a course. By the end of a course, learning outcomes clearly specify what learners are expected to know, perform, apply, and value after completing the subject. These work as a road map for authorities, teachers and learners



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Let us now discuss learning outcomes (LOs) in detail.

- Learning outcomes are clear, specific and accurate.
- These evaluate learning through clear and well-defined questions. These are objective in nature and have no subjectivity and ambiguity.
- Learning outcomes are measurable in nature.
- Learning outcomes are learner-centric, as they are framed from the learner's perspective, highlighting what the learner is able to do, apply, and perform rather than focusing on others.
- Curricular goals are assessed through learning outcomes.
- The expected behavior and skills of the learner are assessed mostly through action verbs.



CHECK YOUR PROGRESS 6.2

1. What are the advantages of writing learning outcomes in a mathematics classroom? Explain using examples of your classroom teaching.
2. Which of the following is reflected by the National Curriculum Framework (2003)?
 - (a) A Major goal is to eliminate fear of mathematics among students.
 - (b) Learners should work individually.
 - (c) Mathematics learning should be examination-oriented.
 - (d) Learners should be punished for committing errors in mathematics.

Answers: 2. (a)

6.3.6 Examples Of Learning Outcomes

Examples of learning outcomes at the preparatory stage from the curricular goal of Counting and Number Recognition may be framed as follows:

- The learner will be able to count a given number of objects (up to 20) and represent their number correctly.
- The learners will be able to read the given number correctly and draw the same number of seeds from the box.
- The learners will be able to read the given number correctly and tell its predecessor of that number with reason.

- The learners will be able to read the given number correctly and tell the successor of that number with reason.
- The learner will be able to represent the given numbers through Dienes Blocks.
- The learners will be able to compare two given numbers and put the appropriate sign (" $>$ ", " $<$ ", " $=$ ") between them.

At the preparatory stage, examples of learning outcomes related to the curricular goal of the four fundamental operations may be framed as follows:

- Learners will be able to add (subtract/multiply/divide) two numbers with concrete objects or tally marks.
- Learners will be able to solve word problems (involving any of the four fundamental operations) from daily life accurately.

Examples of learning outcomes at the preparatory stage from the curricular goal of shapes may be framed as follows:

- The learner will be able to identify the given shapes (rectangle/square/triangle/circle) and name them correctly.
- The learner will be able to tell the properties of the given shapes (rectangle/square/triangle/circle) in terms of their edges and corners correctly.
- Learners will be able to draw closed and open figures with reason for the same.

Examples of learning outcomes at the preparatory stage from the curricular goal of measurement may be framed as follows:

- Learners will be able to measure the length of a tabletop, a mathematics textbook, or a copy through a scale correctly.
- Learners will be able to estimate and classify the weight of a pencil or a copy or a book or a pencil box or a lunch box as more than half a kg or less than half a kg.

Examples of learning outcomes at the preparatory stage from the curricular goal of data handling may be framed as follows:

- Learners will be able to draw simple information from the given bar graph and histogram.
- Learners will be able to draw bar graphs and histograms using the given data.
- Learners will be able to tell the use of bar graphs and histograms.

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Examples of learning outcomes at the preparatory stage from the curricular goal of Pattern may be framed as follows:

- The learners will be able to identify and tell if a unit is repeated in the given pattern.
- The learners will be able to extend a given number pattern.
- The learners will be able to extend a given geometric (having different shapes) pattern.



ACTIVITY 6.2

1. Go through the mathematics textbook of class I and frame learning outcomes of the first three lessons.
2. Go through the mathematics textbook of class II and frame the learning outcomes of the last three lessons.



CHECK YOUR PROGRESS 6.3

Write four characteristics of learning outcomes in classroom teaching.

6.4 IMPORTANCE OF PLANNING

You would agree with the comment “Planning of any task, smaller or bigger, plays a pivotal role in its success.” The reason being, while planning, the planner pays attention to each detail, challenges and constraints of that task before actually conducting it. It increases the chances of its success to a greater extent. Teaching mathematics is a challenging task, as we need to align the learning expectations of the course with learners of different experiences, interests, abilities, cultures, etc. We need to take decisions regarding classroom management, teaching strategies and approaches to learning, learning resources and technology to be adopted for diverse learner needs. In such situations systematic planning is in aligning the goals of learning a specific subject with the needs of the learner by managing time and resources effectively.

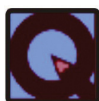
Let us discuss some important points of planning in mathematics.

- While planning the teaching of mathematics, first of all we define **the goals, objectives and learning outcomes of the course** as a whole and then define the same for each lesson specifically.
- On the basis of the goals and objectives, we plan **teaching strategies and engaging activities** to be adopted in the classroom to cater to the needs of different learners. Planning not only helps us in dealing with anticipated

challenges of the class but also in managing learners' behavior for the effective teaching-learning process.

- Proper planning of teaching empowers us for **effective time management to conduct various activities** in the class and ensures that **each topic of the course is delivered within its time**. It provides the learner with various **opportunities to learn and practice skills with a lot of oral work in mathematics**. Time management helps us in sharing learners' experiences in the class.
- Well-planned lessons help us in **creating a positive learning environment in the classroom**. Clear goals and a participatory approach with due emphasis on motivation and feedback to the learner make the teaching and learning of mathematics joyful and interesting for learners.
- Planning enables us to **use various techniques of assessment that align with the learning goals and learning outcomes of the lesson**.
- A well-planned lesson can help us in delivering the lesson in a more effective way which gives us a better experience of managing and teaching the learners. It improves our **self-confidence in the skill of teaching**.
- Planning is a collaborative effort, where teachers of the same subject and class share their experiences and best practices while planning, which **creates a supportive and learning environment for the learners**.

To sum up, one can say that planning in mathematics is about creating a learning, joyful, dynamic and active environment in the class, which is a good factor for both teacher and learner. So, it is a must and an important step of the teaching-learning process.



CHECK YOUR PROGRESS 6.4

1. How does the planning of mathematics help in building a learning environment in the classroom?
2. Which of the following is not a characteristic of effective mathematics planning?
 - (a) Clearly defined learning objectives.
 - (b) Focus on rote memorisation of formulas.
 - (c) Integration of real-world applications.
 - (d) Use of diverse assessment methods.

Answers: 2. (b)



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6.4.1 Types Of Mathematics Planning

Mathematics planning ranges from yearly plans to lesson plans. Each type of planning has its own objectives and focus. Let us start with the yearly plan first.

Yearly Planning

As you know, yearly planning is done just before the beginning of a new session. Lessons of the course are arranged hierarchically based on the psychological principles, maxims of teaching and time schedule. Sequencing of the content/topics is sometimes arranged according to three or four terms of the learners, with the focus that each term has some facts, concepts, procedures and application of the content. At the foundational and preparatory stage, we take care that each term covers some concepts on numbers, geometrical thinking, data handling, etc., comprehensively to avoid monotony and boredom for learners.

Sometimes a yearly plan is given by the authorities, like in schools of the KVS, NVS, Directorate of Education, the Municipal Corporation of Delhi, etc. The plan is formed by some of the senior teachers along with experts in the field and disseminated among all teachers according to their class. This practice brings uniformity to the course and evaluation system. For better yearly planning of mathematics, feedback on the previous year's course design and its execution is also taken care of in order to avoid the challenges faced earlier. This planning provides opportunities to the learner for project work, field visits related to the course, the use of technical resources, hands-on activities in the mathematics lab, peer tutoring, group work, self-learning by the learners, etc. It emphasizes the holistic approach to learning as emphasized by the New Education Policy 2020 (NEP, 2020).

There can be various formats for the yearly plans; for example, on the basis of terms or terminal examination of that class, a month-wise plan or we may develop it according to our needs. There is no fixed format for the yearly plan.

Following is an exemplar format of the yearly plan for class 3, mathematics.

Yearly Planning (2025-26)

Textbook: Mathematics Mela,

Class: Third

Table 6.2 Exemplar Yearly Teaching Plan for Class 3 Mathematics

Term	Lesson No	Lessons Name	Teaching Method	Assessment	Project Work										
First	1	What’s in a Name?	Experiential Learning ● Inter-disciplinary art/sport integrated ● Discovery-based ● Focus on Developing numeracy skills ● Play and activity-based teaching method	Formative Play Way Activity-based 360° evaluation	Write name of each Family member and observe the name with the largest and smallest number of alphabets.										
	2	Toy Joy			Make an object of your choice with three boxes of different shapes										
	10	Fun at Class Party			How will you measure the length of your tabletop?										
Second	3	Double Century			Represent the following numbers through bundles of matchsticks: 129 245										
	4	Vacations with Nani Ma			Frame a question for the following:125 + 78										
	5	Fun with Shapes			Choose any three shapes and make a rangoli on a grid paper										
	6	House of Hundreds-I			Make a game of Ludo using the operation of “+,” and “-” on numbers from 1 to 100.										
Third	7	Raksha Bandhan			Make a list of arrangements you do on the occasion of Raksha Bandhan in your family.										
	13	Time Goes On			With the help of the annual calendar, calculate the number of days you have a holiday in your school.										
	8	Fair Share			Color the following as given: <table><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table> One-half: Red One-third: Green One-sixth: Yellow										

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Term	Lesson No	Lessons Name	Teaching Method	Assessment	Project Work
	14	The Suraj Kund Fair			Describe the local fair you have ever visited and draw three things you saw there.
	9	House of Hundreds-II			There are 850 rubber bands in a box. In how many ways can you count them?
Fourth	11	Filling and Lifting			List five things in your classroom that are heavier than 1 kg.
	12	Give and take			Solve the following using a number line: $140 + 84$ $215 - 197$

In the above format, we may include the number of periods to be devoted to each lesson. Sometimes we may mention the activities to be performed in the class for each lesson. There can be many variations in the format according to our needs and resources available in the class.



ACTIVITY 6.3

- Go through the textbook of mathematics for class 4 and develop a yearly plan term-wise for the same according to resources available in your school.
- Go through the textbook of mathematics for class 2 and develop a yearly plan month-wise for the same according to resources available in your school.



CHECK YOUR PROGRESS 6.5

- How many types of mathematics planning are there? What is the difference between them?
- Yearly planning of mathematics is done by _____.
 (a) Principal (b) Authorities
 (c) Learners (d) Teachers

Answers: 2. (d)

We have discussed the yearly plan till now. The other two types of planning are **unit plans** and **lesson plans**. Before we start the discussion on unit plans and lesson plans, we should have a clear idea of the curricular goals of a subject. The understanding of curricular goals and learning outcomes of a unit will help us in better planning of a unit as well as a lesson.

6.5 UNIT PLANNING

While discussing different types of planning in mathematics, we discussed that there are three types of planning namely, yearly, Unit and lesson planning. We have discussed the format of yearly planning in previous pages of this lesson. Now, we will discuss unit planning.

In a mathematics textbook we find there are many chapters or lessons in the book. You will observe that some lessons have common objectives; such lessons together make a unit in mathematics. In this way we can say that a unit in mathematics consists of related concepts. Any book of mathematics may have many units. For example, here follows the unit plan of the mathematics textbook (Maths Mela) for class III by NCERT.

Here in the first unit, “What’s in a Name,” the numbers from one to one hundred are discussed. Then various methods of counting and grouping are discussed through activities and learning material. So, we can say that unit planning of the content helps us in deciding the sequence of concepts within a unit, support material, and technique of assessment to be used in that unit.

6.5.1 Need of Unit Planning

You may think, what is the need for unit planning, or why can’t we teach without a unit plan? The answer is unit planning brings significant improvement in the quality of the teaching-learning process in the classroom. Let us clarify how unit planning is useful for us in making teaching meaningful and effective:



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Fig 6.2 Math Mela, Class 3 (NCERT, 2023, content page)

- While planning a unit, we consider the entire unit as a whole and recognize the interconnectedness of its components. It helps us in organising the structure of the unit with time and resources available at our disposal.
- It helps us design the content of the unit in a sequential, graded and systematic manner, which acquaints us with developing classroom activities in an effective way.
- It helps us plan multiple or alternative approaches to teaching and learning, which can cater to the individual differences of the learner.

- It helps in planning the unit-wise assessment of learners.
- It helps us in organising remedial teaching as well as enrichment programs as per the needs of the learners.
- It helps us give a harmonious arrangement to various elements of the unit under reference.

6.5.2 Steps of Unit Planning

We have seen that the primary aim of a unit plan is to ensure meaningful and effective learning. As planners, this involves two key tasks: selecting the content and arranging it in a logical sequence. While making these decisions, factors such as the time available for each unit and the nature of the content play an important role. Care must be taken to avoid imbalance—for instance, one unit being too large and taking over a month to complete, while another is so brief that it finishes in just a few periods. With this in mind, let us consider some guidelines that can be followed for effective unit planning:

- List the complete course or subject content or competencies to be achieved by the learners of that class.
- Estimate the total time available for completing the subject content.
- Arrange the subject content or competencies in a proper and logical sequence on the basis of psychological principles of learning.
- Identify the relation and linkages between various subject content and competencies.
- Formulate the course content or competencies in different units on the basis of the following principles:
 - A unit should neither be too small nor be too lengthy.
 - A unit should have elements of commonness within its components.
 - It should not require a period of more than a month to complete a unit to avoid boredom for the learner.
 - It should be arranged in such a manner that a feeling of accomplishment and satisfaction develops for the teacher and the learner.
 - It should be arranged in such a manner that it may further be divided into lesson plans.
 - There is no hard and fast rule regarding the number of days required to complete a unit, but generally we prefer that a unit be completed within 20 days and there should not be more than 10 competencies in any unit.

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6.5.3 Development of A Unit Plan

A unit plan may be developed in various ways, as there is no fixed format for the same. The content area or the competencies to be covered in each unit must be planned. A suggestive format of unit plans is given below:

Unit Plan No.:

Class:

Subject:

Duration:

Name of the teacher:

Area: Understanding Numbers up to 100 (Part 1)

Competencies Covered (Course content covered): Here, we mention the course content or competencies to be achieved. Some exemplary competencies are mentioned below:

2.1.1 (Counting of objects in groups)

2.1.2 (making a grid of 100 units using ten strips)

2.1.3 (Regrouping of numbers in tens and ones)

2.1.4 (Comparing quantities up to 100)

2.1.5 (Estimate the number of objects and verify through counting objects)

2.1.6 (Recognising numbers as labels)

Unit: Write name of the unit

Lesson: Write name of the lesson and mention the lesson number

Pedagogy: The pedagogy to be adopted for the transaction of the unit may be mentioned here. NEP, 2020 is suggesting the play way method; activity-based method; demonstration method; experiential learning, etc. at the foundational and preparatory stages. For example:

Teaching Learning Material: Depending upon the content/competencies, we use teaching learning material for the mentioned unit. Some exemplary teaching and learning material for the competencies mentioned above are as follows:

- Unit and ten cards,
- Objects like seeds, leaves, beads, etc. for counting
- Cubic rods
- Dienes blocks

Learner Activities: Activities performed by the learners are written here. For example they may be involved in some hands-on-activities or group activities or peer learning or interaction among peers or problem solving or joyful activities or experiential learning, etc.

Assessment: The assessment technique adopted by the teachers is written here. It may be through rubrics or some project or problem-solving or through some activities, etc. NEP 2020 focuses on 360° assessment.



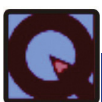
ACTIVITY 6.4

1. Choose any topic from the class III mathematics textbook and develop a unit plan on that.
2. Choose any topic from the class IV mathematics textbook and develop a unit plan on that.

6.5.4 Limitations of Unit Plan

A unit plan is very important and useful for effective teaching-learning, but it has some limitations also. Let us discuss these limitations one by one.

- A unit plan gives a partial view of the course content or competencies developed during a year.
- It is sometimes difficult for the planner to anticipate the teaching method and number of days to be devoted for completion of the unit beforehand.
- Sometimes the division of content or competencies gives artificial views.
- It reduces the flexibility and spontaneity of the teacher if the plan is followed rigidly.
- All the teachers may not be able to plan a good unit plan.



CHECK YOUR PROGRESS 6.6

1. How is a unit plan important for teachers?
2. Which of the following is not a part of the unit plan in mathematics?
 - (a) Clearly defined learning outcomes.
 - (b) A variety of engaging activities.
 - (c) A detailed daily schedule.
 - (d) Appropriate assessment methods.



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3. Which of the following is not a benefit of unit planning in mathematics?
- (a) Gives teachers a holistic view of curriculum.
 - (b) Ensure a systematic arrangement of content.
 - (c) Supports differentiated instructions in the classroom.
 - (d) Allows for a rigid adherence to a predefined teaching approach.

Answers: 2. (c) 3. (d)

6.6 LESSON PLAN

A lesson plan is a plan for teaching a lesson in one period in advance. It helps teachers to foresee the problems related to the lesson in advance that may arise during lesson delivery. It also helps teachers to transact the focus points, linkages, and comparisons of the content of the lesson logically and meaningfully. Thus, we can say that a lesson plan is a method of taking decisions about the selection, sequencing, delivery and execution of various activities to be performed during the class in order to provide learning experiences to the learners.

6.6.1 Need of Lesson Planning

A lesson plan is a must irrespective of the subject or the class to be taught. Lesson planning helps us in avoiding situations like getting stuck during teaching a class, finishing teaching before the period is over, or having some important elements of the lesson excluded from the lesson. Thus, we can say that lesson planning helps us in the following ways:

- It makes teaching organised, systematic and effective.
- It helps in appropriate selection and sequencing of the content for the lesson.
- It helps us to utilise the time available for the lesson properly.
- It helps us in developing insights about learners' needs, abilities and achievements.
- It helps us in planning activities that strengthen learners' participation in the lesson. Thereby making teaching a two-way communication.
- It strengthens our self-confidence in teaching.
- It helps us in the identification and remediation of learning difficulties of the learners as assessment is done along with teaching.

6.6.2 Steps of Lesson Planning

- While developing a lesson plan, first of all we select the content for teaching. Content should be interesting and meaningful for learners. It should be relevant in the day-to-day life of the learner. Thus, the first step of the lesson plan is **selection of the topic or content**.
- The second step is **the formulation of general objectives and specific objectives** for teaching that content. Specific objectives are formed keeping in mind the competency to be developed in learners while teaching that content.
- The third step is **the selection of an appropriate method of teaching** that content. The method is chosen according to the nature of the content and the needs of the learner, which provide active participation of the learner through hands-on activities.
- The fourth step involves **the selection of appropriate techniques of assessment** in order to evaluate the lesson being delivered.

The above steps are shown through the following flowchart:

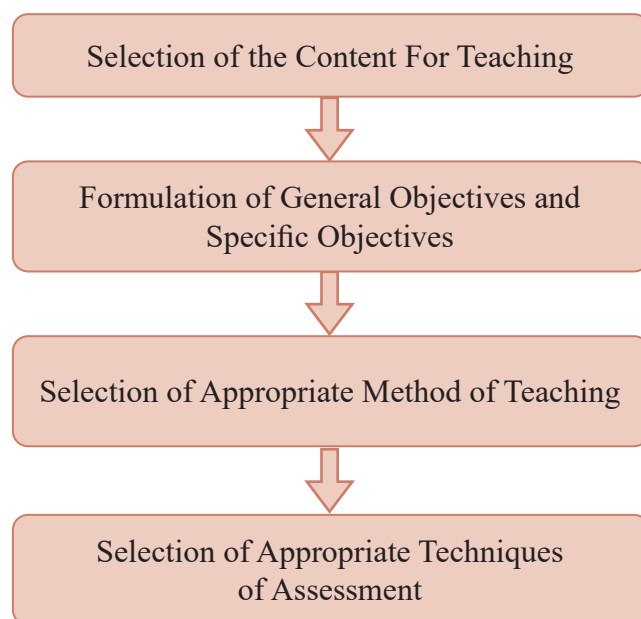


Fig 6.3 Flowchart Illustrating Steps of Lesson Planning

6.6.3 Development of A Lesson Plan

How to teach a lesson may be planned in various ways. Usually, the Herbartian method of lesson planning was used for this process. In the constructivist approach of teaching, learners actively participate in the teaching-learning



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process and construct learning on their own. It is a very structured approach. So, we use a constructivist lesson plan based on the 5 Es (Engage, Explore, Explain, Elaborate and Evaluate) approach, which involves the following five steps:

1. **Engage:** This phase aims to engage learners in the content being taught by some interesting tasks, like asking some puzzles, giving problems to solve, demonstrating some activity, giving learners a task, or showing some pictures or video clips to them. It should link their previous knowledge to the new concept.
2. **Explore:** Learners are given opportunities to explore the concepts through hands-on activities or experiments. It helps them to discover and verify ideas and concepts either independently or in small groups.
3. **Explain:** The teacher explains the concepts through the introduction of the key words and new ideas, which helps learners in constructing their own learning.
4. **Elaborate:** Learners apply their learning to new situations, deepen their understanding, and use the concept for real-world applications.
5. **Evaluate:** This phase evaluates the learner's understanding and fosters reflection. It may include quizzes, projects, activities, problem-solving tasks, or other methods that showcase their learning.

Example: A 5Es Lesson Plan on Counting in Groups

Class: II

Subject: Mathematics

Lesson: I (A Day at the Beach)

Date: -----

Time: 40 Minutes

Name of teacher: -----

Learning Objectives:

After going through this lesson, the learner will be able to:

1. Count objects in groups of 10.
 2. Give examples of objects that come in groups of 10.
- **Engage:** Start by showing a picture of a fruit seller (selling bananas, oranges, chikoos and watermelon) and ask learners to speak out what they are observing in the picture. Ask them the names of fruits they are observing. Ask them to count the number of different fruits they have observed.

- **Explore:**

Ask learners how they have counted the number of different fruits. They may be asked whether they have counted the number of fruits one by one or in groups. Ask them which method is easier and faster: counting one by one or in groups.

****Give learners cut-outs of boxes, each containing ten pencils or ten pens. Ask them to count the total number of pencils and pens in two or three such boxes. Guide them to discover that counting in groups of ten makes the process easier.**

- **Explain:**

The teacher explains how counting 27 pencils, 18 pens, 32 beads, etc., is easier in groups of ten. The teacher takes a sufficient number of examples and shows that: 25 pencils = 2 boxes and 5 pencils; 32 beads = 3 mala and 2 beads; 45 chikoos = 4 Trays of 10 chikoos and 5 chikoos, etc.

- **Elaborate:**

Learners will apply their knowledge by solving word problems involving counting objects in groups of 10. For example:

Eshan has gone to his father's stationery shop today to help him with his work. Three customers came to buy pencils. Help Eshan choose the correct number of boxes and loose pencils by filling in blanks in the given table.

Customer	Number of pencils purchased	Number of Boxes	Number of pencils
1	23	-----	-----
2	39	-----	-----
3	45	-----	-----

- **Evaluate:**

- Learners will be given a quiz that assesses their understanding of counting in groups.
- They may be asked to name three objects that come in groups of tens.
- They may be given a project to name three things that come in groups other than 10 as homework.

* Here instead of learning outcomes, competencies may also be taken.

**** For learners with special educational needs (hearing impairment, visual impairment, and learning difficulties).**



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**ACTIVITY 6.5**

1. Develop a lesson plan on Fun with Shapes for class III.
2. Develop a lesson plan on parts and wholes for class IV.

6.6.4 Limitations of Lesson Planning

Although lesson planning is very important for teachers, as it provides them with the opportunity to think about the sequence of events and anticipation of probable problems in advance, it still has some limitations.

- Too much emphasis on lesson planning may make classroom teaching a routine.
- It is difficult to anticipate all types of classroom situations in advance. So, sometimes it lacks relevance.

**CHECK YOUR PROGRESS 6.7**

1. Which of the following is the most important aspect of making a lesson plan for mathematics?
 - (a) Listing classroom rules.
 - (b) Organising instructions, activities and content systematically.
 - (c) Recording learner attendance.
 - (d) Assigning grades to learners.
2. Which of the following reflects a limitation of lesson planning in contemporary instructional practice?
 - (a) It may restrict teacher flexibility by encouraging rigid adherence to predetermined steps.
 - (b) It enhances the teacher's ability to anticipate and manage diverse classroom situations.
 - (c) It ensures continuous classroom relevance regardless of learner variability.
 - (d) It promotes thoughtful sequencing of learning activities.

Answer: 1. (b) 2. (a)

6.6.5 Conversion of A Unit Plan into Lesson plans

A unit plan may be converted into lesson plans by breaking down the broader unit goals and content into smaller lessons on a daily basis. This process requires careful conversion of the unit's objectives, content, and assessments into logical sequencing in order to make daily lessons engaging and effective for learners.

The following steps may be followed for the conversion of a unit plan into lesson plans:

1. **Understanding Unit Plan:** Goals and learning outcomes of the unit are defined clearly and objectively.
2. **Analysing Content:** The content covered in the unit, including key concepts, topics, operations, facts, formulae and skills, is Analysed in meaningful chunks.
3. **Aligning Assessment:** Align the types of assessments (formative and summative) planned for the unit with the learning objectives.
4. **Breaking units into lessons:** The above three points will help in breaking units into lessons by choosing appropriate content with objectives, learning outcomes and assessment techniques.
5. **Sequencing the Content:** The content of the unit is arranged logically and the content to be covered in each lesson is decided.
6. **Formulate Lesson Objectives:** From the objectives and learning outcomes of the overall unit, we formulate specific, measurable objectives for different lessons.
7. **Designing activities:** On the basis of learning objectives, plan engaging and interactive activities for each lesson that support and provide opportunities for practice and application of the content.
8. **Selecting resources:** Choose appropriate resources, such as textbooks, articles, videos, learning material or online materials, to support the lesson activities and learning.
9. **Connecting lesson to prior knowledge:** Connect each lesson planned to previous lessons and build upon new concepts using previously acquired knowledge in order to create a cohesive learning experience.



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**6.7 LET US SUM UP**

It is important for teachers to appreciate the need and importance of planning before actually teaching. Planning helps the teacher to think in advance about the sequence of teaching and learning activities and to organize classroom interaction meaningfully.

Planning can be done for the whole year and is then known as yearly planning. In yearly planning, the curricular goals of the course or subject are considered so that the teacher gains a broad understanding of what is to be taught throughout the academic session.

Planning may also be done for a unit, which is called unit planning. While preparing a unit plan, due emphasis is given to learning outcomes or competencies that students are expected to achieve. Unit planning helps teachers arrange related lessons together in a systematic and meaningful manner.

A plan can also be made for a single lesson of about 40 to 50 minutes, which is known as lesson planning. Lesson planning helps in organizing teaching in an efficient and effective way. Therefore, it is important for teachers to develop the habit of planning units and lessons regularly.

Lesson planning developed on the 5E's approach (Engage, Explore, Explain, Elaborate, Evaluate) is learner-centered and encourages students to participate actively in the teaching-learning process and construct learning on their own.

**6.8 END EXERCISES**

1. How does a unit plan help in building a learning environment in the classroom?
2. According to you, when should a teacher of mathematics do unit planning for the course?
3. Write the steps of the 5E's lesson planning on "shapes" for class III.
4. Justify the statement, "Lesson planning is a roadmap for teachers."

6.9 REFERENCES

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